OCTO-EARTH

PROJECT REPORTS

Orbital relocation concept, launcher and power-system calculations, climate mirrors, and reinforcement-learning evaluation

EXECUTIVE SUMMARY
OCTO-EARTH is a hypothetical simulation in which an eight-arm launcher ejects asteroid material so the Earth–Moon system recoils from a 1 AU orbit toward 2.7 AU. A solar-collector belt beams power to the launcher, while a space-mirror system supplies additional sunlight. A PPO controller selects launch and climate actions. The evidence supports a functioning simulation and one successful saved-checkpoint test; it does not establish real-world engineering feasibility, climate safety, or robust reinforcement-learning performance.

Project objective

The model targets a semi-major axis of 2.70 AU (±0.05 AU), eccentricity ≤0.05, inclination ≤5°, and a 10,000-simulated-year deadline. It also tracks Moon retention, Earth–Mars separation, a 288 K climate target (±5 K), and finite mass and energy budgets. These are model acceptance criteria, not a proposal to move Earth in reality.

System at a glance

Subsystem	Modelled role	Key declared scale
E1 — launcher	Ejects reaction mass; Earth recoils by momentum conservation.	8 arms; up to 10^{18} kg per group packet; up to 5,000 km/s; 8 launches/year; 50% kinetic efficiency.
E2 — solar power belt	Collects sunlight and beams microwave power to arm-tip rectennas.	5 million collectors at 0.5 AU; assumed 30% of solar output intercepted; 5.8 GHz beams.
E3 — mirror ring	Reflects additional sunlight onto Earth at 2.7 AU.	1.6×10^{15} m² film area; 90% reflectivity; terminator orbit at 20 Earth radii.
PPO controller	Chooses launch throttle, direction, speed and mirror-light set-point.	31-value observation; 5 continuous actions; decisions every 100 years.

How to read this report: Pages 2–4 summarize physics, launcher structure/thermal loads, and solar power/mirror calculations. Pages 5–7 describe the simulation architecture, RL evidence, and mission comparison. Page 8 records limitations and recommended validation.

1. Mission physics and order-of-magnitude calculations

The concept relies on conservation of momentum. When a packet of asteroid material is ejected in one direction, the remaining Earth–Moon system recoils in the opposite direction. Repeated, carefully directed impulses can change orbital energy and angular momentum. The model assumes the asteroid is docked to Earth and treats Earth, Moon and remaining reservoir as a combined system for its primary estimates.

1.1 Mission constants and targets

Quantity	Value used	Meaning
Solar gravitational parameter, μ	1.32713×10^{20} m³/s²	Approximate Sun–system orbital parameter.
Initial / target orbit	1 AU → 2.7 AU	Target distance from the Sun.
Earth–Moon–asteroid system mass	6.0756×10^{24} kg	Model mass for the orbital screen.
Mission horizon / control interval	10,000 years / 100 years	At most 100 policy decisions.
Asteroid reservoir	3.00×10^{22} kg	About 0.408 Moon masses and 0.502% of Earth mass.

1.2 Circular-orbit speed screen

For a circular orbit around the Sun, $v = \sqrt{(\mu/a)}$. At 1 AU, $v_0 \approx 29{,}784.7$ m/s. At 2.7 AU, $v_t \approx 18{,}126.4$ m/s. The supplied slow-spiral screen is $\Delta v = v_0 - v_t \approx 11{,}658$ m/s. A Hohmann-transfer reference is approximately 10,997 m/s, but its two large burns do not represent the gradual launcher strategy.

1.3 Orbital energy and power

The circular-orbit energy change is estimated by $\Delta E = \mu M(1/(2a_0) - 1/(2a_t))$, giving about 1.70×10^{33} J. Over 10,000 years ($T \approx 3.16\times10^{11}$ s), $P = \Delta E/T \approx 5.38\times10^{21}$ W—roughly 30,982 times the sunlight currently intercepted by Earth.

1.4 Launcher mass, speed and energy

The model sets a maximum packet mass of 10^{18} kg, a maximum launch speed of 5,000 km/s, and eight group launches per year. With launch efficiency $\eta = 0.50$, the minimum-energy screen uses the entire 3.00×10^{22} kg reservoir at a launch speed around 2.355×10^{6} m/s. Estimated launch energy is 1.66×10^{35} J, averaging 5.27×10^{23} W over 10,000 years. At the 5,000 km/s cap, the estimated minimum transfer time is 1,771 years under maximum packet rate and speed.

Screening metric	Reported value	Interpretation
Earth escape speed	≈11.2 km/s	Small relative to assumed thousands of km/s launch speed.
Packet kinetic energy at 2,480 km/s	$\approx3.07\times10^{30}$ J	Per 10^{18} kg group packet.
Power drawn per packet at 50% efficiency	$\approx6.15\times10^{30}$ J	Remaining half is loss/heat in the model.
Peak launcher power at 5,000 km/s	$\approx6.34\times10^{24}$ W	Enormous planetary-scale power demand.

“Feasible” means the idealized mass, speed, energy and timing constraints pass a model gate. It does not mean the launcher, asteroid docking, power network or Earth-safe operation can be built.

2. Eight-arm launcher: structural and thermal screening

The calculation insert assumes a packet follows a curved guide with radius $R^c = 3R_e$. The guide's centripetal force is $F = mv^2/R^c$. It assumes eight equal load paths and an ideal circular load-bearing cross-section for each arm. These assumptions are critical; this is not a completed structural design.

2.1 Static stress equation and radius ambiguity

With total load shared equally across eight arms, $F_{allo}w = 8\pi r_{arm}^2 \sigma_{allo}w$. Equating it to $mv^2/(3R_e)$ gives $v_{max} = \sqrt{(24\pi\sigma_{allo}w\, r_{arm}^2 R_e/m)}$. Using $\sigma_{allo}w = 86.67$ GPa, $r_{arm} = 3.0\times10^6$ m, $m = 10^{18}$ kg and $R_e = 6.371\times10^6$ m gives $v_{max} \approx 1.9357\times10^7$ m/s (19,357 km/s). The assumed allowable corresponds to a 130 GPa ultimate strength divided by safety factor 1.5.

Case	Total radial load	Per-arm load	Mean stress / assumed area
Operating cap: 5,000 km/s	1.308×10^{24} N	1.635×10^{23} N	5.78 GPa (6.67% allowable)
Equation limit: 19,357 km/s	1.960×10^{25} N	2.451×10^{24} N	86.67 GPa (100% allowable)

Stress scales with v^2 under this simplified model. Crucially, if "3,000 km" means arm reach/length rather than cross-section radius, substituting it for r_{arm} is invalid.

2.2 Waste heat per launch

Quantity	Calculation / assumption	Reported estimate
Packet kinetic energy	$\frac{1}{2}mv^2$; $m=10^{18}$ kg, $v=5\times10^6$ m/s	1.25×10^{31} J
Energy drawn at 50% efficiency	Ekin/η	2.50×10^{31} J
Waste heat per event	Edraw − Ekin	1.25×10^{31} J
Per arm per event	Equal split across 8 arms	1.56×10^{30} J
Average heat at 8 events/year	Annualized, not pulse power	3.17×10^{24} W total; 3.96×10^{23} W/arm

This is a system-level loss estimate, not proof that all heat enters the arm. If fraction f_{arm} is deposited locally over pulse duration t_p, a first pulse-power estimate is $f_{arm}(1.56\times10^{30}\text{ J})/t_p$. Neither parameter is supplied, so actual temperature rise cannot be calculated.

2.3 Thermal stress and combined failure checks

For a uniformly heated member whose expansion is fully restrained, $\sigma_{th} \approx E\alpha\Delta T$. A freely expanding member has much lower uniform axial thermal stress, but gradients, constraints, joints and rapid heating/cooling can cause bending, local stress and fatigue. The report's illustrative example (not the specified arm material) uses $E=200$ GPa and $\alpha=12\times10^{-6}$/K, giving 2.4 MPa/K; 100 K implies 0.24 GPa and 500 K implies 1.20 GPa.

Before a safety claim, specify actual cross-sectional area and curvature, material strength versus temperature, elastic modulus, thermal expansion, conductivity, heat capacity, emissivity, heat-deposition fraction, pulse duration, radiator geometry, joints, buckling, stress concentrations, fatigue and creep.

3. Solar power belt, microwave transfer and climate mirrors

The energy chain is sunlight → thin-film collectors → electricity → 5.8 GHz microwave transmission → rectennas at the eight arm tips → launcher. The mirror ring is a separate climate subsystem that reflects sunlight onto Earth when its orbit is farther from the Sun.

3.1 Collector belt and microwave chain

Parameter	Model value / result	Explanation
Collector belt	5 million collectors at 0.5 AU	Assumed to intercept 30% of solar output; construction and station keeping idealized.
Sunlight intercepted	≈1.15×10^{26} W	Model's 30% solar-output assumption.
Microwave transmitter efficiency	80%	20% transmitter loss in energy ledger.
Capture fraction	99.88% at 1 AU; 86.30% at 2.7 AU	Calculated belt geometry and blockage assumptions.
Rectenna diameter / count	1,000 km / 8	Each area ≈7.85×10^{11} m².
Rectenna conversion efficiency	85%	Delivered electricity: 1.29×10^{25} W at 1 AU and 1.11×10^{25} W at 2.7 AU.

At the target orbit, modeled electrical output is about 1.76 times maximum launcher peak demand. This is an idealized aggregate comparison, not proof that power can be routed, phased, received and dissipated safely at this scale. The report estimates 5.30×10^{34} J transmitter heat and 5.48×10^{33} J beam spill over the tested heuristic mission.

The calculation report lists a 114 m/s microwave photon-momentum side effect using $\Delta v = E/(cM)$, but says it was not applied to simulated dynamics. Momentum transfer from absorbed radiation should be accounted for consistently; the listed figure is not an implemented Earth velocity change.

3.2 Collector station keeping and manufacturing

The belt assumes sail-tilt station keeping and factories using Mercury-derived material. One calculation run reports 723 km maximum slot deviation with station keeping versus 1.46×10^{8} km without it. The updated PPO report separately summarizes about 386 km maximum deviation, RMS 317 km, and roughly 51 m/s/year control Δv. These are different run summaries and should not be merged as one identical measurement.

3.3 Mirror capacity and climate model

Quantity	Calculation	Result
Mirror film mass	1.6×10^{15} m² × 0.005 kg/m²	8.0×10^{12} kg
Effective area at 45° tilt	Aeff=A cos45°	1.13×10^{15} m²
Mirror capacity at 2.7 AU	Aeff × 0.90 × 1361 / 2.7^2	≈1.90×10^{17} W
Extra power to restore sunlight	(1 − 1/2.7^2) × 1.74×10^{17} W	≈1.50×10^{17} W
No-mirror equilibrium at 2.7 AU	Two-box energy-balance estimate	≈152 K; severe freeze in model

The two-box climate model couples a surface layer to a deep-ocean layer, with albedo increasing as Earth freezes. At 2.7 AU, the model estimates only 13.72% of present sunlight naturally reaches Earth. The mirrors reflect 90% of incident light; the absorbed 10% is radiated away. Mirror pointing, orbital dynamics, light distribution and real climate response are not demonstrated.

Calculation-review caveat: C121 is flagged for a likely missing circular-aperture factor; the review estimates about 1.31% rather than 2.1% for the cited 100 km receiver case. This is a flagged issue, not a resolved redesign.

4. Prototype architecture and how to reproduce the simulation

The README describes OCTO-EARTH as a Python reinforcement-learning prototype with a 3-D browser viewer. Its Overview mode is explicitly not to scale.

4.1 Run procedure and interface

Install dependencies with `pip install -r requirements.txt`, then run `python octoearth_rl.py`. The local viewer opens at `http://127.0.0.1:8800/` or the next free port. Training starts automatically; Ctrl+C stops it while retaining the saved model.

Viewer area	Purpose and outputs
Mission	Start/stop training; choose Live training, Best PPO model, Best hand-coded (heuristic), or No ejection; select Auto/CPU/GPU; inspect progress.
View	Switch Overview/Earth POV, camera tracking, visible layers and screen panels.
Params	Inspect parameters, units and tooltips; apply valid setting changes before training.
Results	Training curves, PPO/baseline comparisons, target pass/fail, decision table, Mars-safety table and resource counts. Rewards are separate from physical outcomes.
Guide / Ledger / Log	Parameter explanations; climate, energy, collector and enabler ledgers; runtime messages.

4.2 Visual modes and interpretation

Mode / key	What it shows
Overview	Sun, Earth, Moon, Mars, asteroid, trajectories and target; not to scale.
Earth POV / V	Close-up of Moon, asteroid, launcher ring/eight arms, magnetic field, solar wind, Hill sphere and Mars.
Power beams / B	Microwave beams at arm tips; brightness follows recorded power.
Collector / C	Close-up of one solar-collector satellite.
Weather & life / W	Climate under mirrors; “show without” displays the modeled no-mirror frozen state.
Space / R / L / Esc	Play-pause; reset; legend; close the open drawer.

4.3 Simulation/control architecture

The technical report describes a configuration and feasibility gate, a Gymnasium environment, a fast engine for PPO rollouts, a Mars safety shield, a higher-fidelity N-body engine for validation, a climate/energy ledger, and a reporting pipeline. The controller does not directly overwrite Earth’s velocity: its action is decoded into packet mass, launch speed, direction in the radial–transverse–normal frame, and climate-light set-point. The engine applies reaction, charges resources, accounts for packet outcomes, advances Mars interactions and returns the next observation and reward.

Failed alternatives in the README: laser push, hydrogen harvesting, superconducting magnetic arms, and a Dyson beam on a reflector held by the arms. These are comparative failure cases, not working enablers.

5. Reinforcement-learning method and recorded training

The control problem is a Markov decision process. One episode is a full mission; each action is held over a 100-year interval, with up to 800 packet-launch events per step. The observation is a scaled 31-value vector and the action is a five-value continuous vector.

5.1 Observation and action design

Observation group	Indices / count	Information
Earth–Moon barycentre	0–5 / 6	Position and velocity, scaled.
Orbit and resources	6–16 / 11	Semi-major axis, eccentricity, inclination, energy gap, Moon state, remaining mass/energy/time and capability estimates.
Previous action	17–21 / 5	Previous normalized controls.
Mars safety	22–26 / 5	Relative Mars position, safe-distance ratio and predicted closest approach.
Climate and mirrors	27–30 / 4	Temperature error, sunlight share and mirror capacity.

Action	Meaning
a0	Ejection throttle; maps to 0–100% packet mass; below 2% threshold means no ejection.
a1, a2	In-plane and out-of-plane thrust direction in rotating RTN frame.
a3	Launch speed mapped linearly from 20 km/s to 5,000 km/s.
a4	Climate-light set-point, requesting 0.5–1.5 times today's sunlight; mirrors add light up to capacity.

5.2 PPO configuration

Setting	Saved configuration
Algorithm / network	PPO, Stable-Baselines3; MLP [64, 64] for actor and critic; tanh activations.
Rollout / batch / epochs	1,000 / 200 / 10.
Learning rate	3×10^{-4}.
Discount / GAE lambda	$\gamma = 0.995$; $\lambda = 0.95$.
Initial log standard deviation / entropy coefficient	−0.5 / 0.0.
Seed / training target	Seed 1; 1,000 episodes; deterministic evaluation every 50 episodes.

The reward combines progress toward target orbital energy and terminal success with penalties for energy, mass, time, eccentricity, inclination, Mars-safety proximity, climate deviation and terminal failure. Later policies conserve resources by applying almost no useful propulsion and miss the deadline, yet their rewards become less negative. This suggests reward-objective misalignment or policy collapse; the logs do not isolate a single code defect.

5.3 What the training artifacts show

Evidence	Observed result	Interpretation
training_log.csv	1,000 stochastic episodes; 94,586 timesteps; 0 successful episodes.	Every recorded exploration episode failed.
Termination reasons	641 deadline; 359 resources_exhausted.	No stochastic training episode completed the mission.
evals.json	20 periodic deterministic evaluations; episode 50 succeeds; episodes 100–1,000 fail.	Early checkpoint success did not persist.
Best checkpoint	Selected at episode 50, 3,790 timesteps; separate deterministic evaluation succeeds on seeds 1001–1002.	Different from stochastic episode-50 rollout, which failed.
Last checkpoint	At episode 1,000, final $a \approx 1.00198$ AU after 10,000 years; energy fraction 0.0017; success=0.	Not suitable as demo model.

6. Mission results and policy comparison

The updated technical report compares no ejection, a hand-designed heuristic and the saved best PPO checkpoint. The procedure reports fast-model results on three seeds, then one high-fidelity N-body mission for seed 2001 and a 200-year ballistic validation. One high-fidelity seed is not enough to establish generalization.

Policy	Fast success	High-fidelity result	Years	Final a / e	Mass used	Energy used
No ejection	0/3	Deadline; validation fails	10,000	1.0000 AU / 0.01593	0%	0 J
Hand-designed heuristic	3/3	Success; validation passes	3,600	2.7000 AU / 0.00889	95.25%	1.757×10^{35} J
Best PPO	3/3	Success; validation passes	7,200	2.6972 AU / 0.00748	97.28%	1.839×10^{35} J

6.1 Safety and climate

Metric	No ejection	Heuristic	Best PPO
Closest Mars pass	0.37211 AU	0.05005 AU	0.04928 AU
Configured safe distance	0.02456 AU	0.02456 AU	0.02456 AU
Mars-shield overrides	0	200	472
Final surface temperature	288.17 K	288.33 K	286.00 K
Maximum climate deviation	2.08 K	2.57 K	4.68 K
Years outside ±15 K band	0	0	0
Moon bound at mission end	Yes	Yes	Yes
200-year orbit validation	Fail: target not reached	Pass	Pass

The best PPO mission reaches a = 2.69719 AU, eccentricity 0.00748 and inclination 0.2616° after 7,200 years. The 200-year coast test reports maximum |Δa| = 0.00282 AU and a bound Moon. Closest Earth–Mars distance is 0.04928 AU (about 7.37 million km), above the configured 0.02456 AU limit. However, 472 shield overrides—2.36 times the heuristic's 200—mean safety performance partly depends on an external guardrail.

6.2 Resource and climate interpretation

Compared with the heuristic, PPO takes 3,600 additional years, uses about 4.7% more propulsion energy and about 2.0 percentage points more asteroid mass. The heuristic's maximum climate deviation is 2.57 K, leaving 2.43 K margin inside ±5 K; PPO's 4.68 K deviation leaves 0.32 K. Both remain inside the wider ±15 K band. Without mirrors, the model estimates severe cooling around 155–157 K at 2.7 AU, so the habitability outcome depends on the hypothetical mirror system.

Primary conclusion: the saved PPO checkpoint succeeds in one reported high-fidelity mission, but the hand-designed heuristic is the stronger current baseline on time, resource use, climate margin and safety-shield interventions.

7. Limitations, calculation-review notes and next steps

The reports distinguish simulation consistency from physical buildability. The calculation review finds most elementary arithmetic consistent under the declared assumptions, but flags specific issues that should remain visible in any submission.

7.1 Calculation and evidence caveats

Issue	Why it matters / treatment
Arm geometry and stress	The 19,357 km/s result assumes a 3,000 km cross-section radius and equal load sharing. If 3,000 km is arm reach/length, the calculation is invalid. No real structural or fatigue design is provided.
Thermal stress	50% launcher efficiency estimates lost energy, not arm temperature. Material, heat distribution, pulse duration and thermal properties are missing, so definitive thermal stress cannot be calculated.
Impulse divided by initial mass (C39)	The calculation review calls this invalid as a delivered Δv test for a changing-mass/recoil system. Retain 11,658 m/s as an ideal orbital screen, not as independently proven by that ratio.
Photon momentum (C81)	The report lists 114 m/s as an indicative beam push but says it was not applied. A consistent momentum ledger is required.
C121 capture fraction	Review flags a likely missing circular-aperture factor: about 1.31% rather than 2.1% for the cited 100 km receiver case. Recompute before using.
Conflicting mission outcomes	The calculation-review PDF and updated RL report do not fully agree on PPO outcomes/override counts. This report uses the updated RL report for PPO comparison and does not merge differing runs.

7.2 Engineering and model limitations

The model assumes a 3×10^{22} kg asteroid can be docked to Earth; packets up to 10^{18} kg can be launched at 5,000 km/s; the launch system survives shock without damaging Earth; collectors intercept 30% of solar output at 0.5 AU; microwave transfer can deliver enormous power; mirrors can be assembled and pointed; and radiators reject waste heat away from Earth. Hardware, launch shock, atmospheric effects, mirror orbit/pointing, light distribution and real climate response are incompletely modelled or unmodelled.

7.3 Recommended next experiments

Priority	Action	Evidence required
1	Diagnose reward collapse; log each reward component on successful and failed evaluations.	Explain why near-zero propulsion can improve return; document reward changes against a hypothesis.
2	Preserve best checkpoint, final checkpoint and heuristic baselines separately.	Automated, versioned comparison on identical seeds.
3	Evaluate PPO and heuristic on 20–30 unseen randomized phases and multiple high-fidelity seeds.	Success rate and spread for time, mass, energy, climate, Mars distance and overrides.
4	Run safe ablations for Mars shield, climate margins and collector availability.	Separate learned-policy performance from guardrail effects.
5	Recompute flagged beam-capture and momentum terms; verify structural and thermal inputs.	Reviewed equations, named material/geometry, heat-flow assumptions and reproducible calculations.

Conclusion

OCTO-EARTH is an exploratory simulation integrating orbital mechanics, finite-resource propulsion, power beaming, climate control, visualization and PPO control. Under declared hypothetical enablers, the selected PPO checkpoint passes one high-fidelity mission and a 200-year stability check. However, no stochastic training episode succeeded and all later periodic evaluations failed; the heuristic remains faster and more resource-efficient. The defensible conclusion is a simulation demonstration—not a feasible engineering plan for relocating Earth.

Source basis: supplied README(1).md, Calculations.pdf (calculation review), and OCTO_EARTH_RL_Report_Updated.pdf (run evidence dated 25 October 2026).